

RoboFinals-IKEA Docker Deployment Guide

Image Name

代码块

```
1 paperc/robofinals:RoboFinals-IKEA
```

1. Environment Requirements

Recommended host environment:

- Ubuntu 22.04 / 24.04
- NVIDIA GPU with a working NVIDIA driver
- Docker Engine
- NVIDIA Container Toolkit
- X11 desktop session

Check that the GPU is visible:

代码块

```
1 nvidia-smi
```

Check that Docker is available:

代码块

```
1 docker --context default ps
```

Check that Docker can access the GPU:

代码块

```
1 docker --context default run --rm --gpus all nvidia/cuda:12.4.1-base-ubuntu22.04 nvidia-smi
```

2. Pull the Docker Image

Pull the image from Docker Hub:

代码块

```
1 docker --context default pull paperc/robofinals:RoboFinals-IKEA
```

Verify that the image exists locally:

代码块

```
1 docker --context default images | grep robofinals
```

3. Configure GUI Authorization

Run the following command on the host machine:

代码块

```
1 xhost +SI:localuser:root
```

Prepare the X11 authorization file path:

代码块

```
1 export DOCKER_XAUTH="${XAUTHORITY:-/run/user/$(id -u)/gdm/Xauthority}"
2 test -f "$DOCKER_XAUTH" && echo "$DOCKER_XAUTH"
```

If `test -f` does not print anything, the Xauthority file may be located elsewhere. Check the current value:

代码块

```
1 echo $XAUTHORITY
```

Then set it manually:

代码块

```
1 export DOCKER_XAUTH=/actual/path/to/Xauthority
```

Note: inside the container, the authorization file is mounted to `/tmp/docker.xauth`. Do not mount it to `/root/.Xauthority`, as this may cause Docker file/directory mount conflicts.

4. Start the Container

Run the following command on the host machine:

代码块

```
1  docker --context default run --rm -it \  
2    --gpus all \  
3    --network host \  
4    --ipc host \  
5    --privileged \  
6    -e DISPLAY=$DISPLAY \  
7    -e XAUTHORITY=/tmp/docker.xauth \  
8    -v /tmp/.X11-unix:/tmp/.X11-unix \  
9    -v "$DOCKER_XAUTH":/tmp/docker.xauth:ro \  
10   -v $HOME/.cache/lightwheel_sdk:/root/.cache/lightwheel_sdk \  
11   -v $HOME/.cache/ov:/root/.cache/ov \  
12   -v $HOME/.cache/Kit:/root/.cache/Kit \  
13   -v $HOME/.cache/NVIDIA:/root/.cache/NVIDIA \  
14   paperc/robofinals:RoboFinals-IKEA
```

After entering the container, the default working directory should be:

代码块

```
1  /workspace/robofinals
```

Confirm with:

代码块

```
1  pwd
```

5. Activate the Runtime Environment

Run this inside the container before executing the IKEA task:

代码块

```
1  conda activate robofinals
```

6. Run the IKEA Task

All commands below must be run inside the container after activating the `robofinals` conda environment.

This image contains one IKEA-specific task: `AssembleTableTask`, a table assembly task where the robot inserts four legs into the corresponding sockets on a tabletop.

Run the task with GUI:

代码块

```
1 python3 robofinals/scripts/teleop/teleop_main.py \  
2 --task_config local_teleop \  
3 --viz kit
```

The config file is located at:

代码块

```
1 /workspace/robofinals/configs/data_collection/teleop/local_teleop.yml
```

It points to the IKEA scene and sets `task: AssembleTableTask` by default. No edits are needed for the default run.

7. Common Issues

7.1 GUI Window Does Not Open

If the log contains errors such as:

代码块

```
1 Authorization required, but no authorization protocol specified  
2 GLFW initialization failed  
3 Failed to acquire IWindowing interface
```

The X11 authorization file is likely not passed into the container correctly. Re-run:

代码块

```
1 xhost +SI:localuser:root  
2 export DOCKER_XAUTH="{XAUTHORITY:~/run/user/$(id -u)/gdm/Xauthority}"  
3 test -f "$DOCKER_XAUTH" && echo "$DOCKER_XAUTH"
```

Make sure the Docker command contains:

代码块

```
1 -e XAUTHORITY=/tmp/docker.xauth
2 -v "$DOCKER_XAUTH":/tmp/docker.xauth:ro
```

7.2 Docker File/Directory Mount Conflict

If Docker reports:

代码块

```
1 not a directory: Are you trying to mount a directory onto a file
```

Do not mount the Xauthority file to `/root/.Xauthority`. Use:

代码块

```
1 -e XAUTHORITY=/tmp/docker.xauth
2 -v "$DOCKER_XAUTH":/tmp/docker.xauth:ro
```

7.3 Asset Pulling Fails

The default IKEA task uses local scene and task backends, so it should not need to fetch assets from the Lightwheel cloud registry.

If you switch to another config that references the cloud registry and see an error similar to:

代码块

```
1 data_not_found:: no registry found
```

Please check:

- Whether the asset name is correct
- Whether the asset type is `objects` or `fixtures`
- Whether the current account has access to the asset

7.4 Slow First Startup

The first startup can be slow. Common causes include:

- IsaacSim / Kit GUI initialization

- RTX / shader cache initialization
- Local asset loading
- USD scene composition
- PhysX / GPU pipeline initialization

The Docker command above mounts the following cache directories:

代码块

```
1 -v $HOME/.cache/ov:/root/.cache/ov
2 -v $HOME/.cache/Kit:/root/.cache/Kit
3 -v $HOME/.cache/NVIDIA:/root/.cache/NVIDIA
```

Later startups are usually faster.

7.5 Common Non-Fatal Warnings

The following warnings are usually not fatal:

代码块

```
1 Seed not set for the environment
2 TGS solver ... may cause noisy velocities
3 Not all actuators are configured
4 Changing backend from 'numpy' to 'torch'
```

If there is no Python traceback and the GUI/task runs normally, these warnings can usually be ignored.